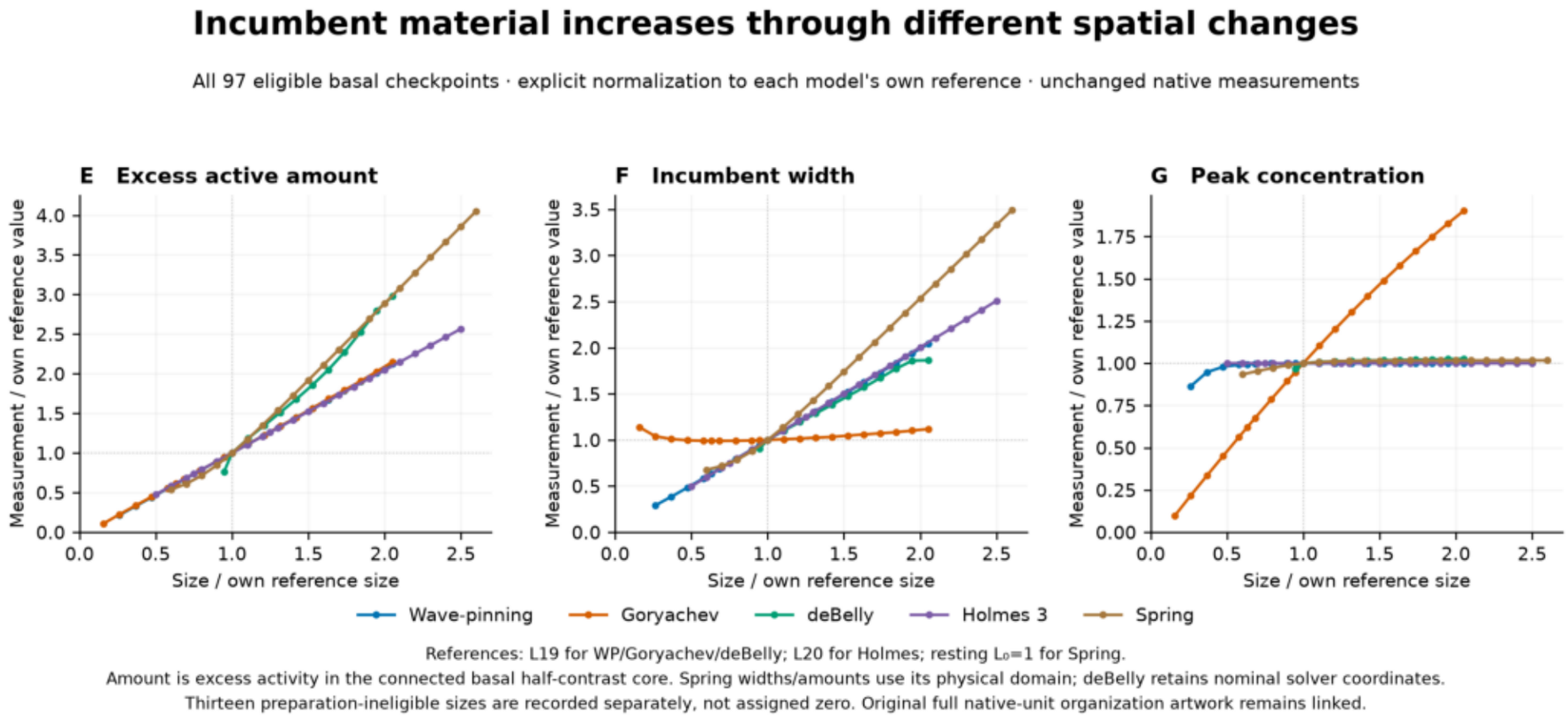
Larger incumbents: the prior all-five foundation

Exact previously retained amount, width and peak comparison; unchanged artwork.



NEW: 72 full-field challenge protocols in WP and Goryachev

Exact polarized basal → independently vary native peak, physical support and pulse duration → observe during cue and release

At both sizes within each model: peak multipliers 0.5 / 1 / 2; physical widths 3 / 5; three common pulse durations.

Readouts: finite original-region activity loss, challenger growth, and retained field response after 10000 cue-free native units.

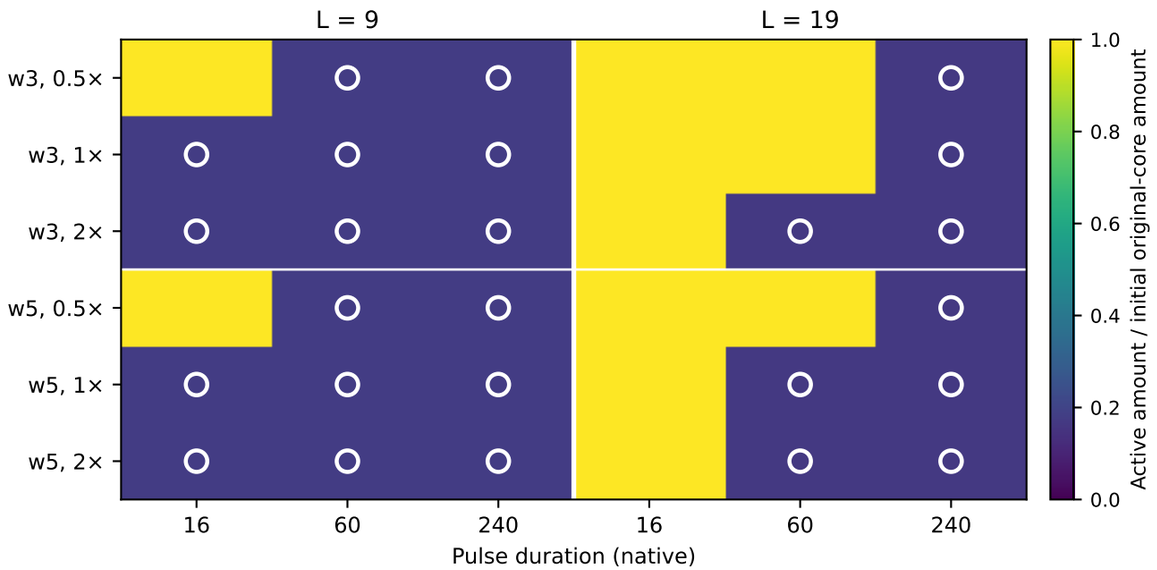
This is a two-model experimental stage, not all-five challenge coverage. Native stimulus × inactive is not a calibrated optogenetic actuator.

The new experiments illustrate consequences of the incumbent organization; they do not uniquely identify molecular mechanisms or allocate causal percentages.

Wave-pinning: performed strength × width × pulse experiment

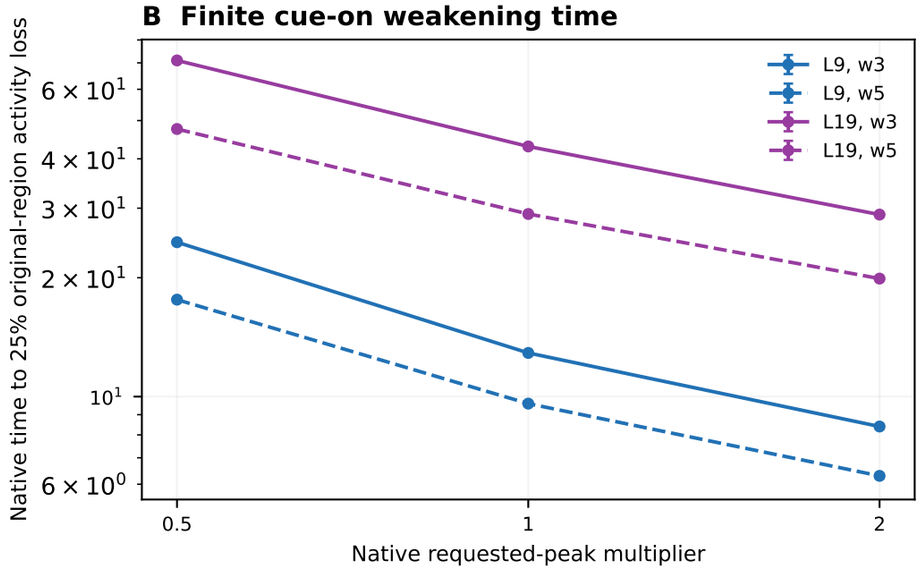
36 protocols; exact shared basals at L9/L19; native requested peak = 0.1 × multiplier.

A Original-core activity after 10000 off units



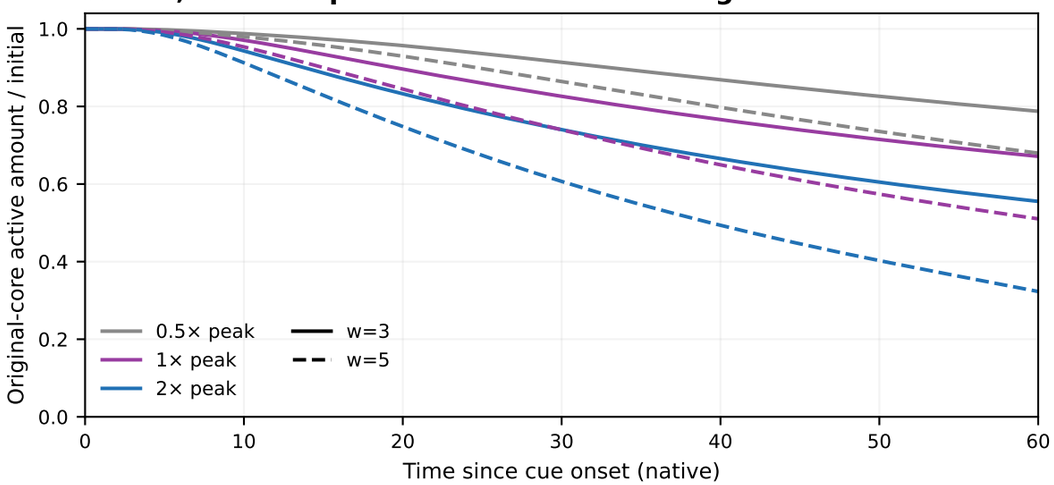
○ White ring: original retained scalar-strength gate passes. Color: measured regional activity.

B Finite cue-on weakening time

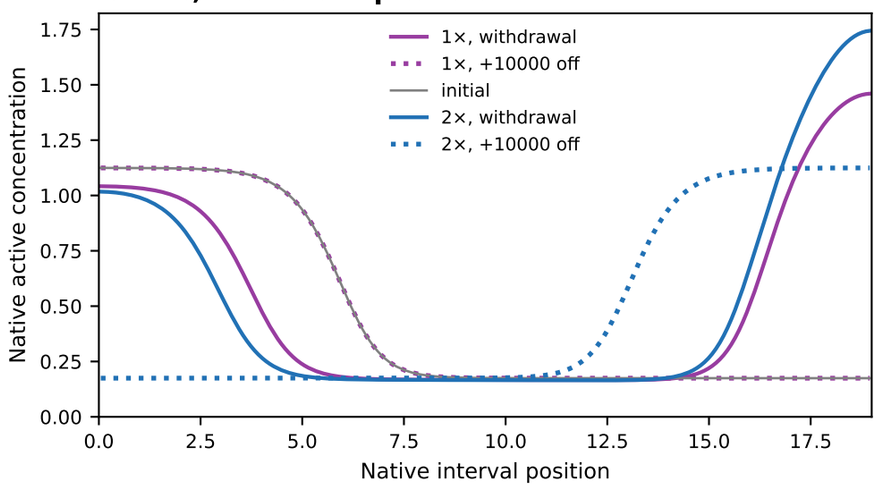


Bars: adjacent saved-time brackets (0.1 unit), not confidence intervals.

C L19, common pulse 60: actual weakening



D L19, w3: actual profiles



At pulse 60, doubling peak or widening to 5 overcomes the tested L19 scalar resistance; L9 passes every tested combination.

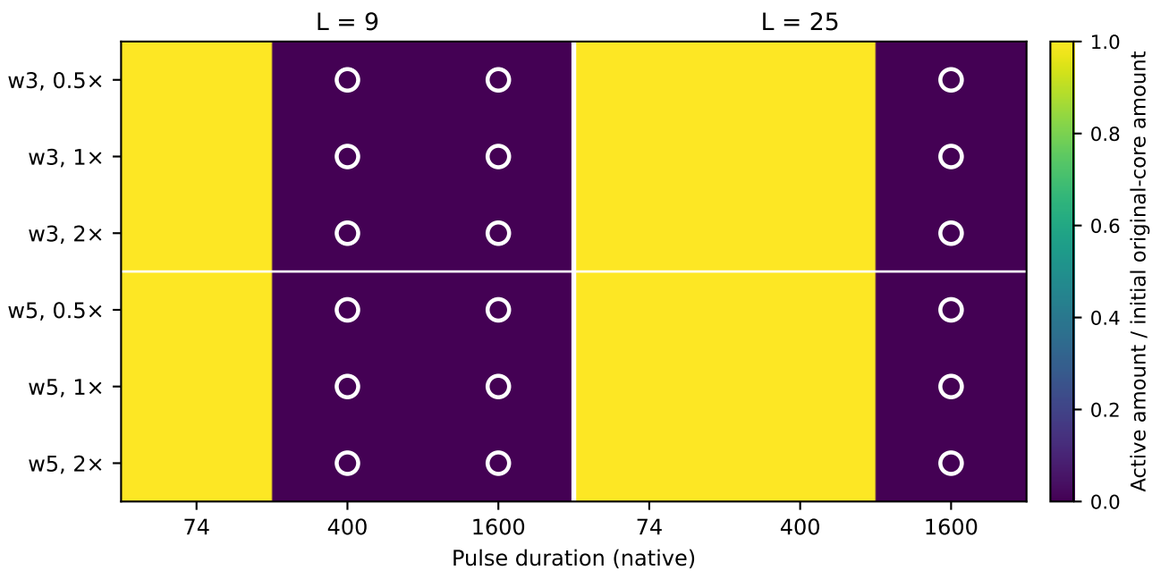
All 36 native field-tail tests pass. Regional loss ≠ identified molecular damage; scalar strength ≠ complete field replacement. N = 201; no spatial-convergence claim.

Native stimulus × inactive and clocks are model-specific. Width changes integrated input. The 25% readout is descriptive; 10%/50% results remain in the source table.

Goryachev: performed strength × width × pulse experiment

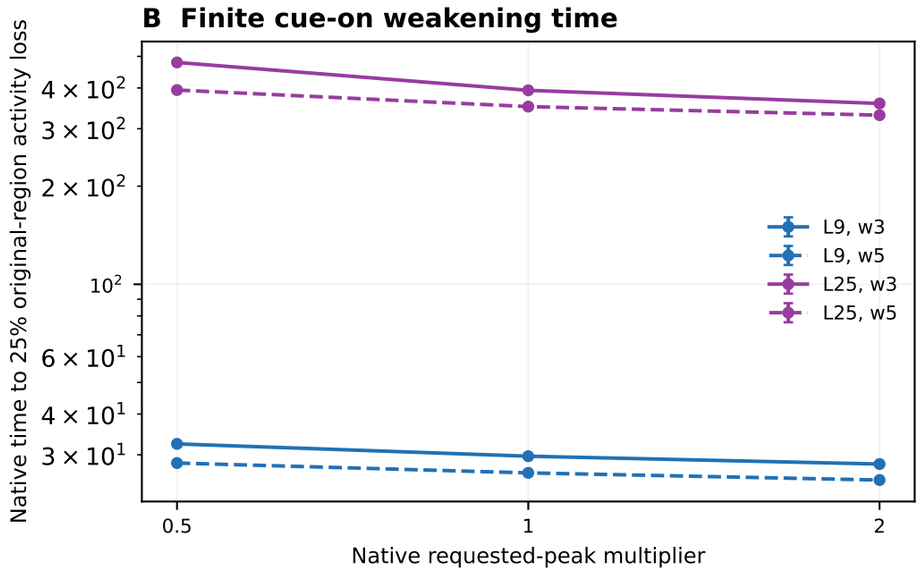
36 protocols; exact shared basals at L9/L25; native requested peak = 6.4 × multiplier.

A Original-core activity after 10000 off units



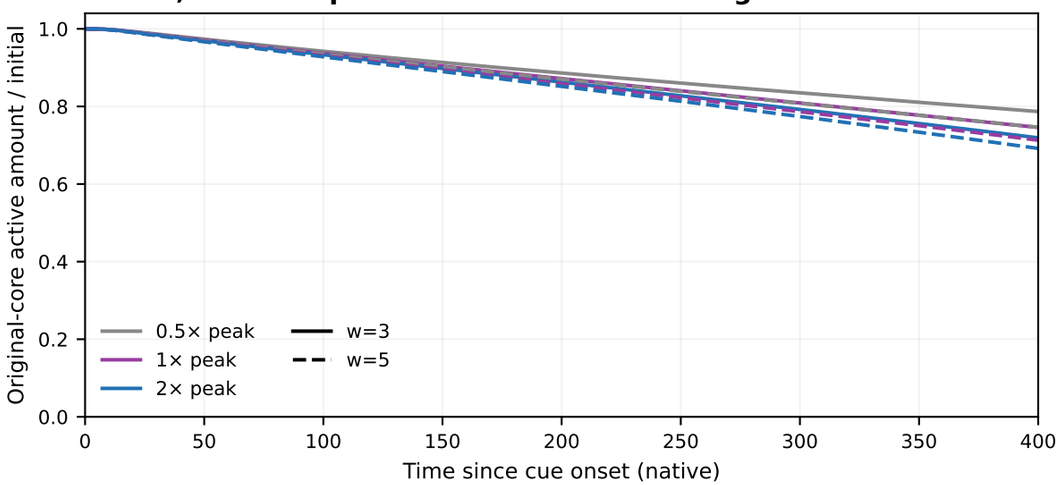
○ White ring: original retained scalar-strength gate passes. Color: measured regional activity.

B Finite cue-on weakening time

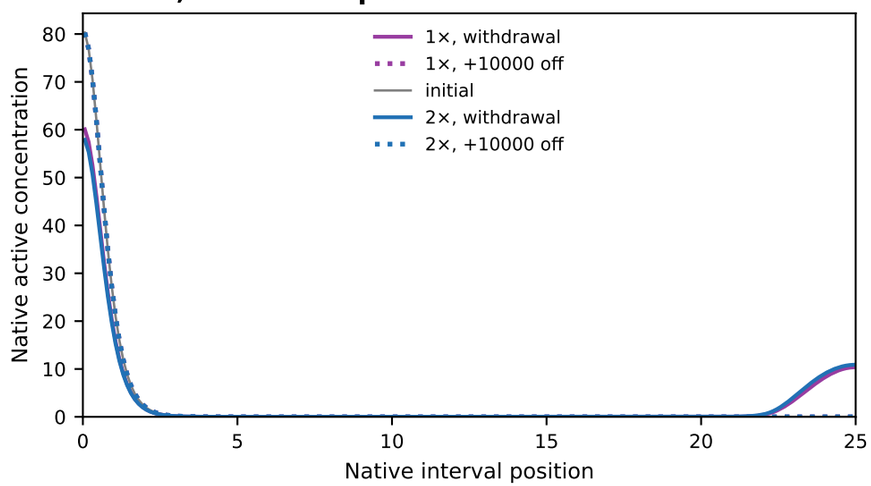


Bars: adjacent saved-time brackets (0.1 unit), not confidence intervals.

C L25, common pulse 400: actual weakening



D L25, w3: actual profiles



At pulse 400, all L9 conditions reverse and all L25 conditions return; pulse 1600 reverses all L25 conditions. Sparse pulses can miss shifted boundaries.

All 36 native field-tail tests pass. Regional loss ≠ identified molecular damage; scalar strength ≠ complete field replacement. N = 201; no spatial-convergence claim.

Native stimulus × inactive and clocks are model-specific. Width changes integrated input. The 25% readout is descriptive; 10%/50% results remain in the source table.